DEPARTMENT OF ACTUARIAL MATHEMATICS AND STATISTICS

HERIOT-WATT UNIVERSITY

F78AA1 Actuarial and Financial Mathematics A

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Feedback on Examination Performance

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General Comments: While some students did well, there were more grades F–D than there were grades C–A. The next semester's course WILL ASSUME MASTERY OF THIS MATERIAL and 3rd year courses make a considerable jump in terms of difficulty and WILL ASSUME MASTERY OF ALL SECOND YEAR COURSES. Students getting a C or less in this examination should regard themselves as being in the danger zone. It is essential to grasp that mathematical courses build progressively on what has gone before and hence each course must not merely be passed but must be MASTERED.

Failings that the poorer performances highlighted were as follows.

- Lack of knowledge of basic mathematics, for example solving a quadratic equation. by trial and error instead of using the quadratic formula. See comments above about mastery of what has been studied previously.
- General lack of precision. Question 3 was intended to test whether a student could take concepts mentioned often in the lectures and formulate precise statements about them.
- Inability to write down proofs and derivations.
- Inability to understand mathematical objects as representations of real-world quantities, for example a symbol $(Ia)_{\overline{n}|}$ as the present value of a precisely specified series of increasing cashflows.
- Related to the preceding point, inability to translate written statements about a real-world problem into corresponding mathematical statements.

Comments on individual questions are over the page.

1. (a) Many students got this correct, but many struggled either to write down the basic definition of $(Ia)_{\overline{n}|}$, or to use basic algebra.

(b) Parts (i) and (ii) were done correctly by most. A common error in (ii) was to forget that £500 p.a. payable half-yearly means £250 per half-year. Part (iii) was very poorly done. Very few spotted that the payments at integer durations define an increasing annuity whose p.v. is straightforward to calculate, and the payments occurring 1/4 year, 1/2 year and 3/4 year before integer durations are copies of that same set of cashflows, just shifted in timing by 1/4 year, 1/2 year and 3/4 year respectively.

2. This question was well done by most, and more students scored full marks here than in any other question.

3. This question was intended to test ability to formulate a precise statement and any vagueness was penalised. Most students got some marks in (a). The most common error in (b) was to express the content of Steffensen's theorem correctly, but in the form of a necessary rather than a sufficient condition.

4. Part (a) was straightforward and was done well. Most students who were able to formulate the equation of value in (b) went on to give a correct answer. In part (c) it was necessary to figure out that CGT was payable; those who did this correctly usually got the right answer. Overall, the weakest answers to this question indicated very scanty attention to the topic during the semester.

5. This was straightforward and the answers given were generally either very good — many students getting full marks — or very poor, showing clearly that the topic had been neglected during the semester. Both equations of value in (a) were quadratic equations and, while many students did apply the quadratic formula, many still tried to solve them by trial and error. A striking error that was made quite often was to formulate the LIRR as the 1/3 power of the accumulation factor over the whole period, instead of the 1/4 power. This suggests memorisation of examples seen in class rather than understanding of the material.

6. Like Q.5, answers were either reasonable attempts or hopeless. The one-year pause before profits began to be earned caused most problems, although a similar example had appeared in class.

A final general comment: many of the weaker scripts showed clearly that attention had lapsed in the second half of the semester, especially in the last two or three weeks. Thus, quite a few attempts at Q.4, Q.5 and Q.6 showed only the vaguest awareness of the subject matter. Some of the topics in the second semester build upon the concepts introduced towards the end of the first semester — see earlier comments about mathematical subjects being progressive — so anyone who slackened off during the latter has some rapid catching up to do.